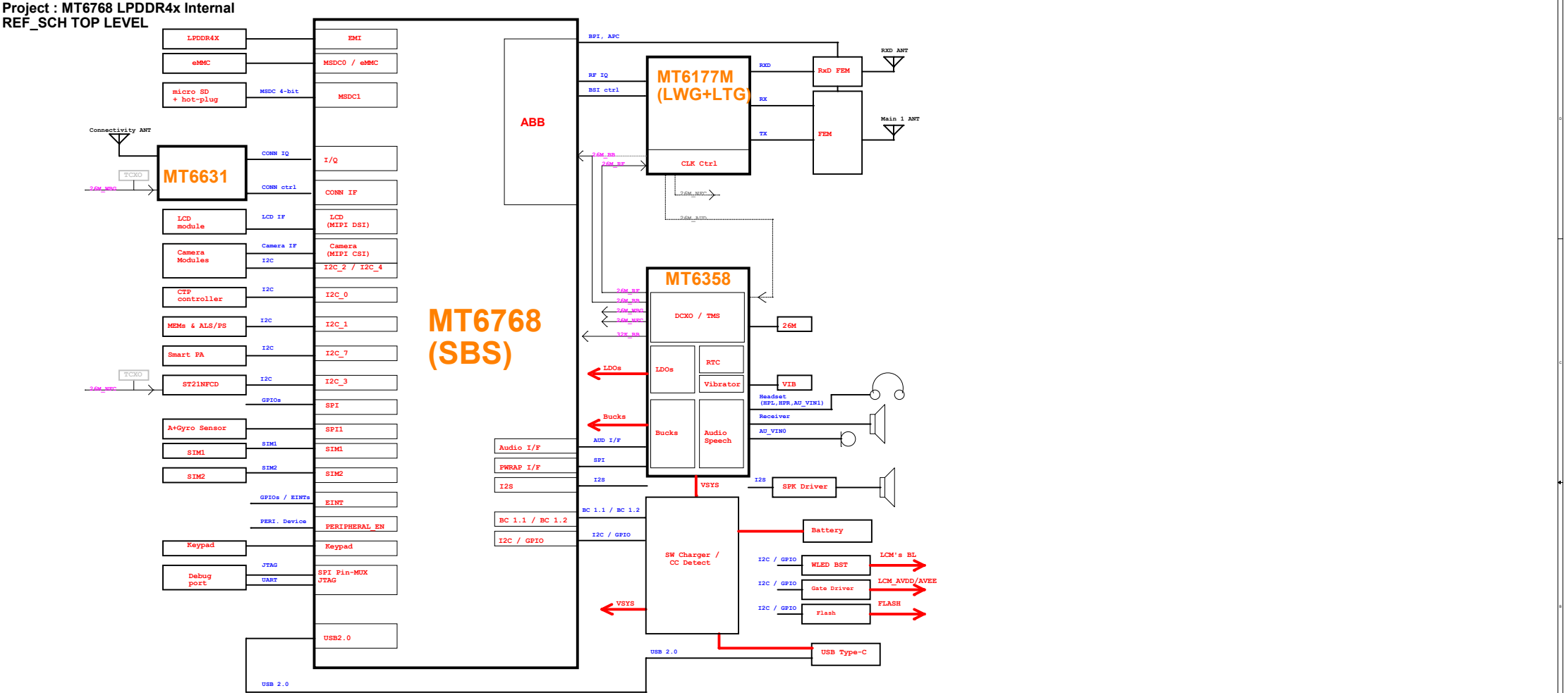




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I2C	Sub SYS	Function	Part Number	I2C Spec.	Budget Timing	I2C Slave Address / Write / Read (7-bit mode)	
I2C-0	AP	Cap Touch controller	GT1151	400 Kbps	Yes.	0x5D	Write:0xBA / Read:0xBB
		LCM Gate Drive	RT4801HWSC	400 Kbps	Yes.		
I2C-1 (I3C)	AP Sensor Hub	Magnetic Sensor	AK09918C	400 Kbps	Yes.	0x0C	Write:0x18 / Read:0x19
		Ambient Light Sensor	CM36558	400 Kbps	Yes.	0x51	Write:0xA2 / Read:0xA3
		Proximity Sensor					
		Pressure Sensor	BMP280	400 Kbps	Yes.	0x77	Write:0xEE / Read:0xEF
I2C-2 (I3C)	AP	Rear Camera	IMX519	400 Kbps		0x1A	Write:0x34 / Read:0x35
			EEPROM	400 Kbps		0x50	Write:0xA0 / Read:0xA1
			AF driver	400 Kbps		0x0C	Write:0x18 / Read:0x19
			SSK2P7	400 Kbps		0x2D	Write:0x5A / Read:0x5B
		Dual cam (B+b) - Main1	EEPROM	400 Kbps		0x50	Write:0xA0 / Read:0xA1
			AF driver	400 Kbps		0x0C	Write:0x18 / Read:0x19
I2C-3	AP	NFC	ST21NFCD	400 Kbps	Yes.	0x08	Write:0x10 / Read:0x11
		Dual cam (B+b) - Main2	OV8856	400 Kbps		0x10	Write:0x20 / Read:0x21
			EEPROM=NA				
			AF driver=NA				
I2C-4 (I3C)	AP	Front Camera	SSK2T7	400 Kbps		0x2D	Write:0x5A / Read:0x5B
			EEPROM	400 Kbps		0x52	Write:0xA4 / Read:0xA5
			AF driver=NA				
I2C-5	AP	Reserved					
I2C-6	AP SSPM	Reserved					
I2C-7	AP	Audio Smart PA	MT6660	400 Kbps	Yes.	0x34	Write:0x68 / Read:0x69
		SW Charger	SW Charger	400 Kbps	Yes.		
		CC Detect	RT1715WSC	400 Kbps	Yes.		
		Flash LED 5V Boost	RT4505	400 Kbps	Yes.		

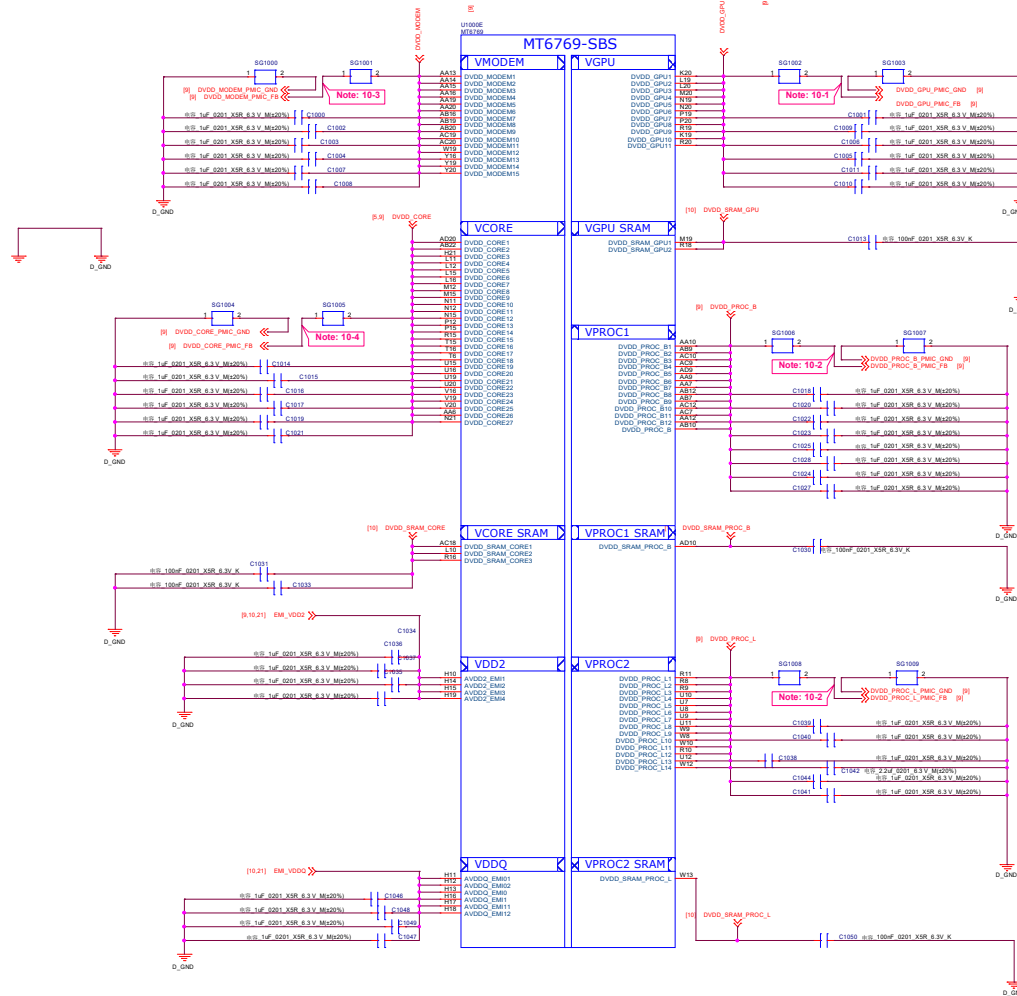
Note : I2C Spec. : Standard mode (100 kbps) and Fast mode (400 kbps), Fast mode Plus (1 Mbps) and High-speed mode (3.4 Mbps)

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Date	Category	Item
2018.12.14 (V0.1)	Page 04	V0.1 Release
2019.01.11 (V0.2)	Page 12	Add CAM_PDN3(GPIO110) HW pin for Note 12-4: HW pin for DDR type feature in bootstrap Delete R1007 and R1012 ext. pull for LP4X eMCP
	Page 13	Add CAM_PDN3(GPIO110)'s Note : No ext. pull for DDR type feature in bootstrap
	Page 69	Add CAM_PDN3(GPIO110)'s Note : No ext. pull for DDR type feature in bootstrap
	Page 21	Add C2106 for VBIF28 cap
	Page 22	Delete R2202

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Schematic design notice of "10_BB_POWER_PDN" page:

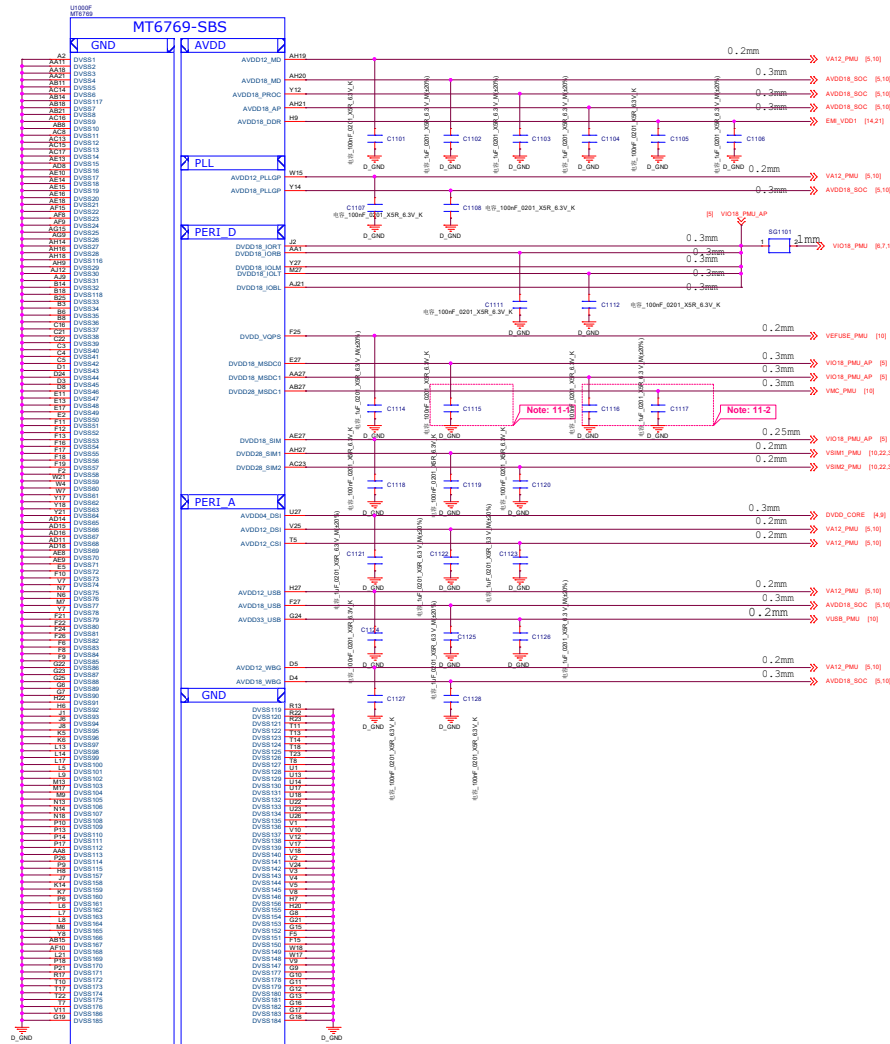


Note 10-1: Differential pair of DVDD_GPU remote sense must be close to BB's ball.
Remote sense trace with GND shielding to PMIC (Differential)

Note 10-2: Differential pair of DVDD_PROC remote sense must be close to BB's ball
Remote sense trace with GND shielding to PMIC (Differential)

Note 10-3: Differential pair of DVDD_MODEM remote sense must be close to BB's ball.
Remote sense trace with GND shielding to PMIC (Differential)

Note 10-4: Differential pair of DVDD_CORE remote sense must be close to BB's ball.
Remote sense trace with GND shielding to PMIC (Differential)



Schematic design notice of "11_BB_POWER_IO" page:

Note 11-1: C1115 closed DVDD18_MSDC0 150mil

Note 11-2: C1116 closed DVDD18_MSDC1 150mil
C1117 closed DVDD28_MSDC1 150mil

AVDD12 : 单个点走8mil, 两路分支各16mil, 总路径20mil

VIO18_FMU_AP: 总路径40mil

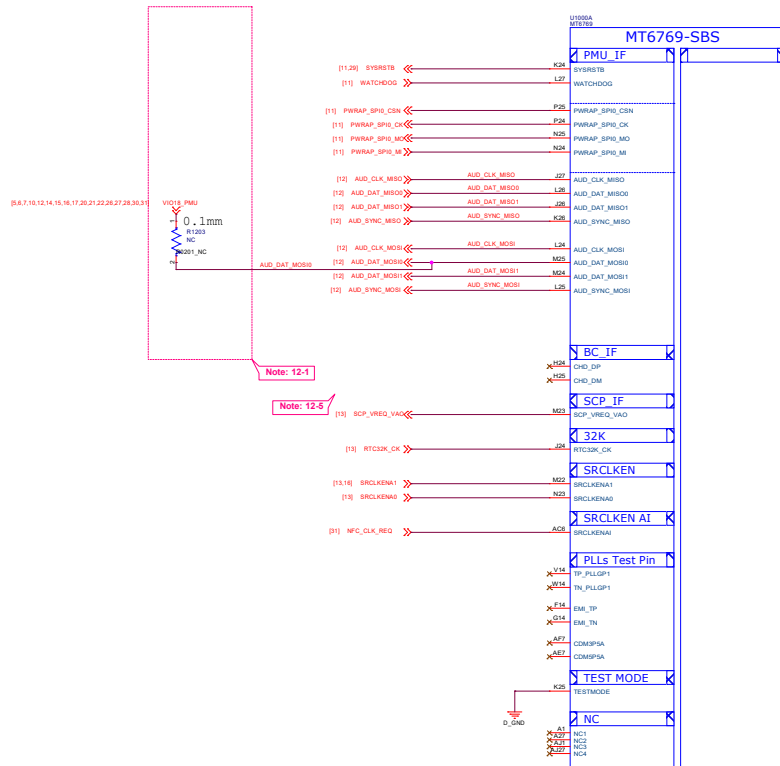
AVDD18_SOC: 总路径20mil

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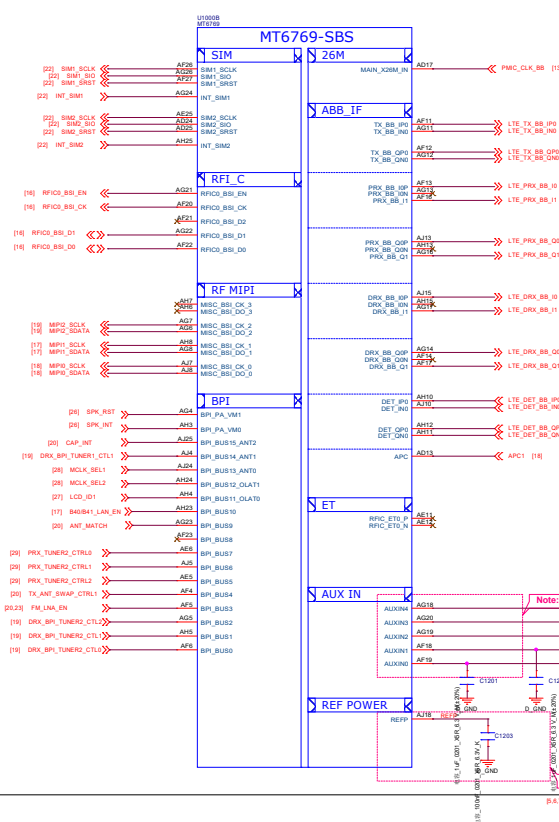
Schematic design notice of "12_BB_1" page:

Note 12-1: PWRAP_SPI0_CSN" and "AUD_DAT_MOSIO" are bootstrap pins to select which interface will be the JTAG pin out

PWRAP_SPI0_CSN	AUD_DAT_MOSIO	JTAG Function	
default=PU	default=PD	AP_JTAG	MD_JTAG
HI	LO	N/A	N/A
HI	HI (by ext. PD)	SPI0 + EINT8	SPI2 + SPI3
LO	LO	SPI0 + EINT8	N/A
LO	HI (by ext. PD)	N/A	N/A

Note 12-2: To shunt a 1uF capacitor in the AUXIN ADC input to prevent noise coupling. It should be placed as close to BB as possible. Connect the unused AUX ADC input to GND.

Note 12-3: The de-coupling cap. for REFP (AJ18 ball) have to be placed as close to BB as possible.



Note 12-4: HW pin for DDR type feature in bootstrap (refer to BB HW design Notice)

AUD_SYNC_MISO	AUD_CLK_MISO	CAM_PDN3	PMIC 6358 voltage	DDR Type
default=PD	default=PD	default=PD	VDRAM1 / VDRAM2	DDR
No ext. pull	No ext. pull	No ext. pull	1.125v / 0.6v	LP4X eMCP
No ext. pull	12K pull to VIO18	No ext. pull	OFF / 1.8v	Reserved
12K pull to VIO18	No ext. pull	No ext. pull	1.225v / OFF	LP3 eMCP
12K pull to VIO18	12K pull to VIO18	No ext. pull	1.125v / 1.8v	Reserved

Note 12-5: Charger must have D+/D- pin for charger type USB detection. Charger must have at least 500mA USB current for All charger type.



Thermistor to sense RF PA temperature

1. R1208 must close to L1E Band 7 PA or the bottom PA ~2mm.
2. The distance is the shortest distance from package edge to edge.

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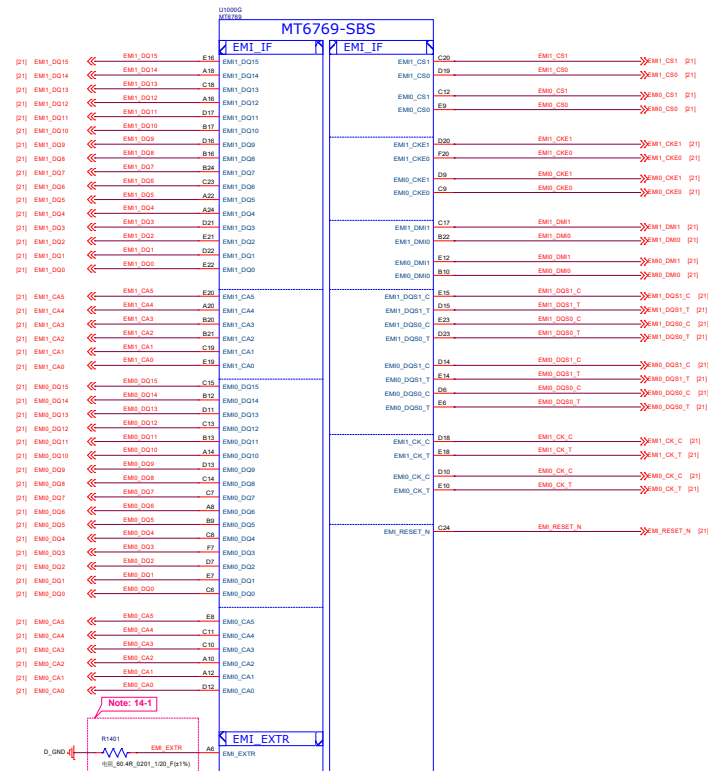
Title 01_Block_Diagram

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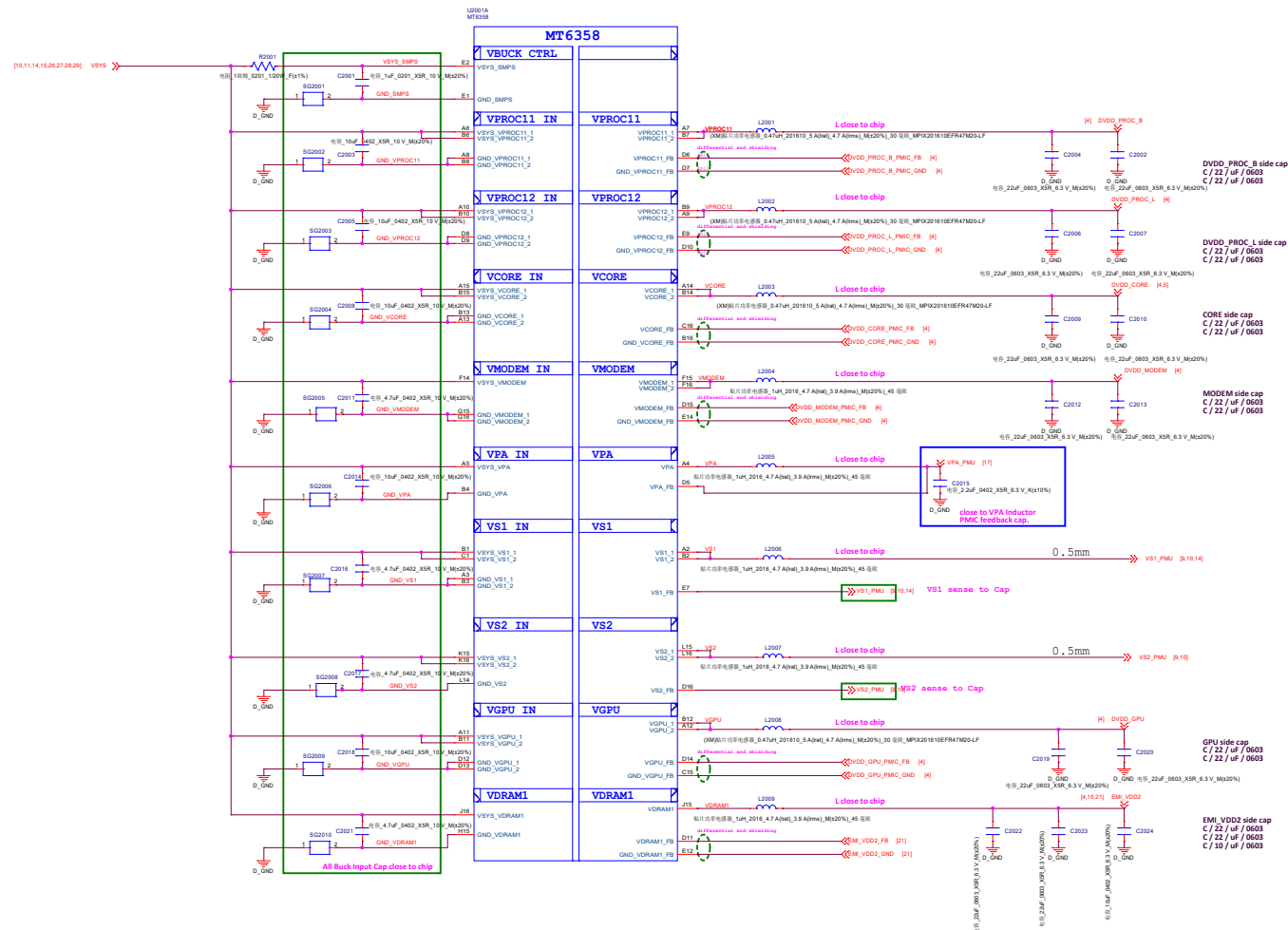
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Schematic design notice of "14_BB_3" page.

Note 14-1: R1401 please select 60.4 ohm (1%) resistor

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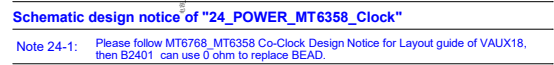
Title 01_Block_Diagram

Size D Project J15_MB_V1

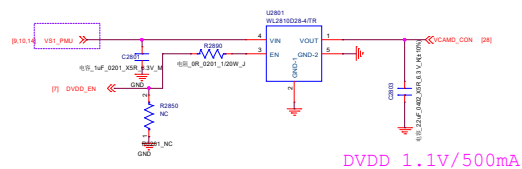
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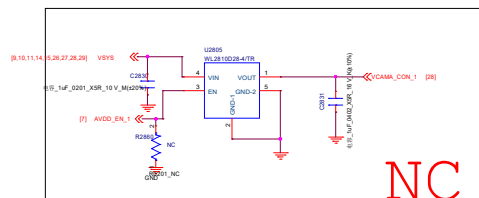


LDO for Rear Camera (VCAMD_EXT1) 1.06V



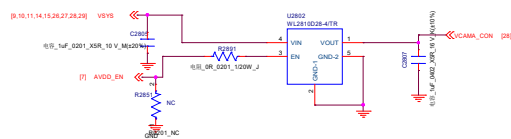
DVDD 1.1V/500mA

LDO for Dual Camera (VCAMA_EXT) 2.8V

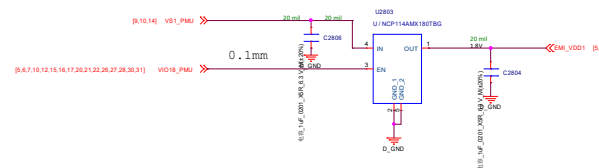


NC

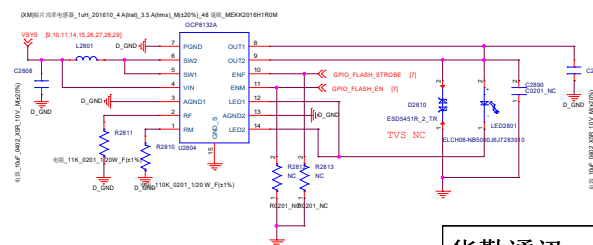
LDO for Dual Camera (VCAMA_EXT) 2.8V



Ext. EMI_VDD1 for LPDDR4X VDD1



Flash LED 5V Boost



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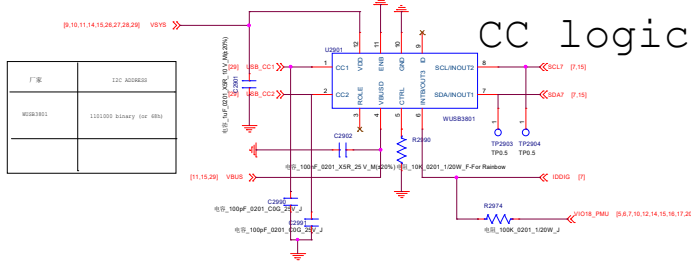
Title 01_Block_Diagram

Size D Project J15_MB_V1

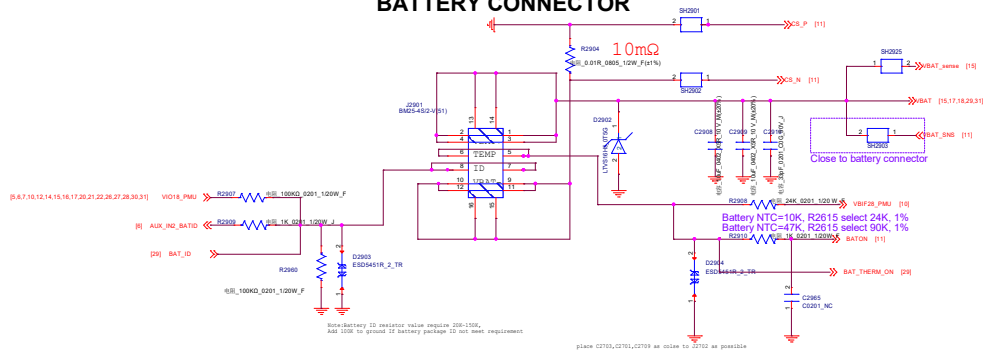
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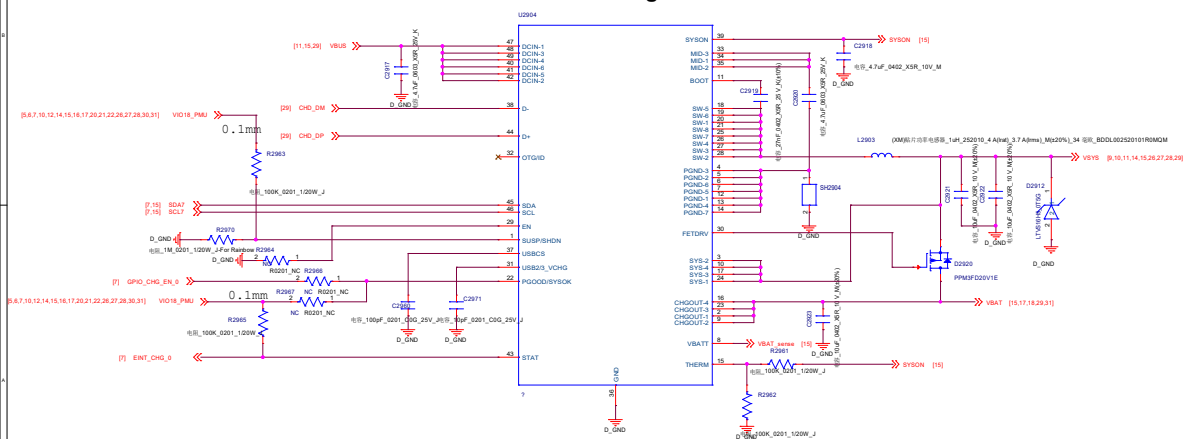
BATTERY CONNECTOR



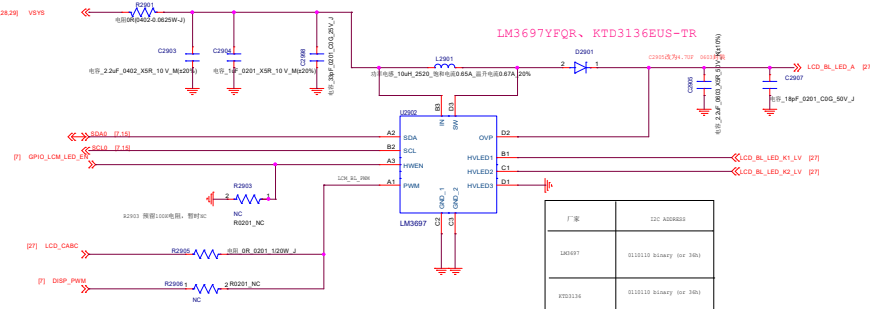
Schematic design notice of "29_EXT_WLED+DB+FL+CHG+PD" page.

Note 29-1: For better ESD or surge performance we need choose suitable device for system protection. Please refer to [Surge device selection guide V2.0] provide by MTK.

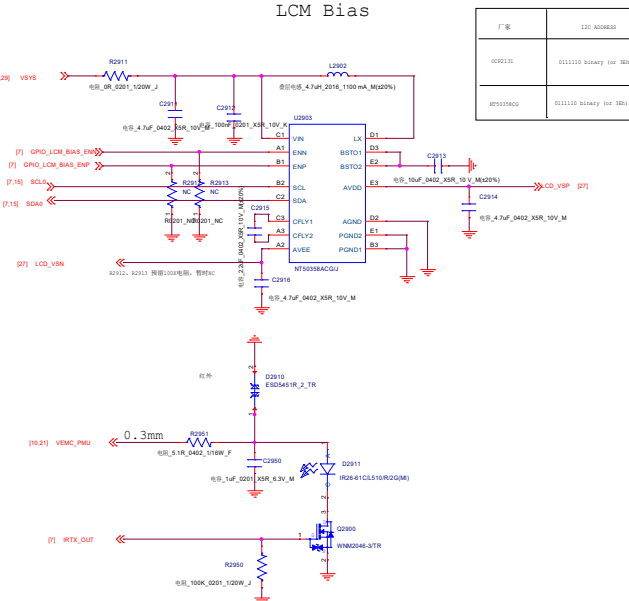
SW Charger



LCM Backlight LED Driver



LCM Bias



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Title 01_Block_Diagram

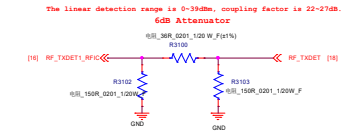
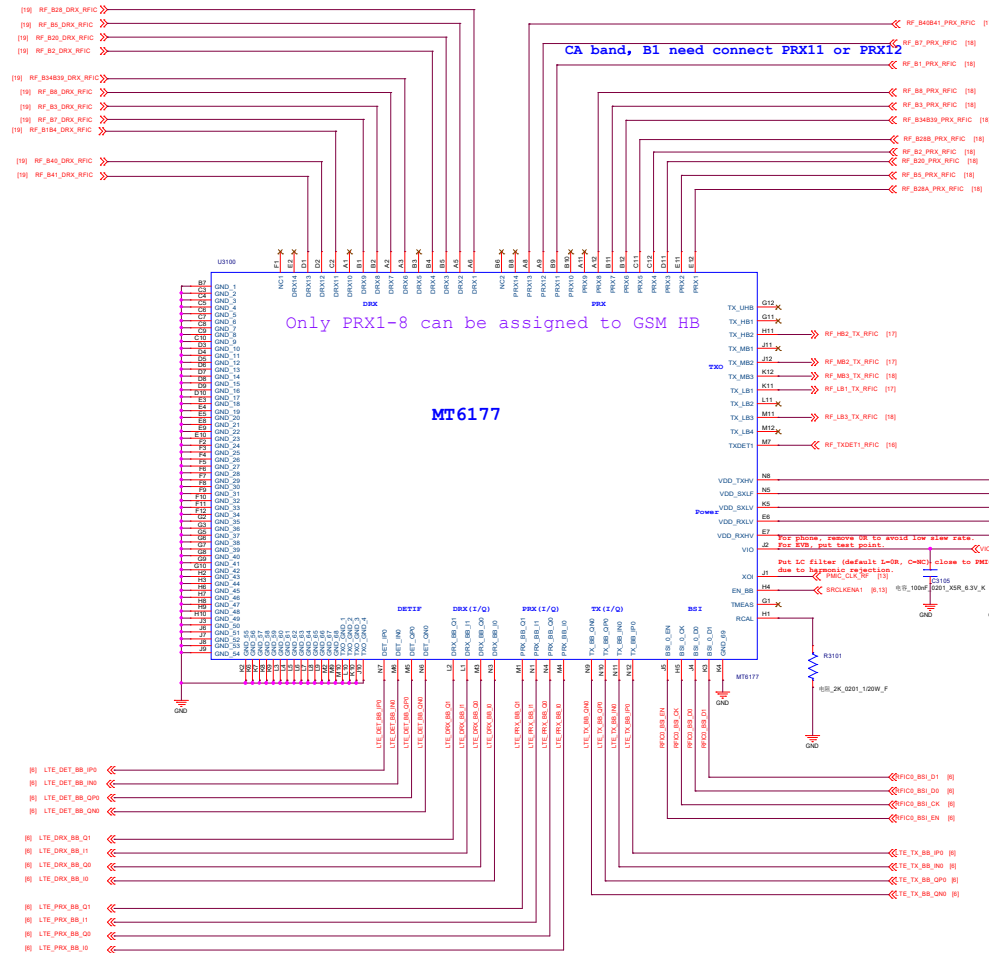
Size D Project J15_MB_V1

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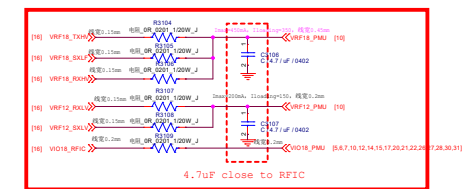
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Attenuator for Detector



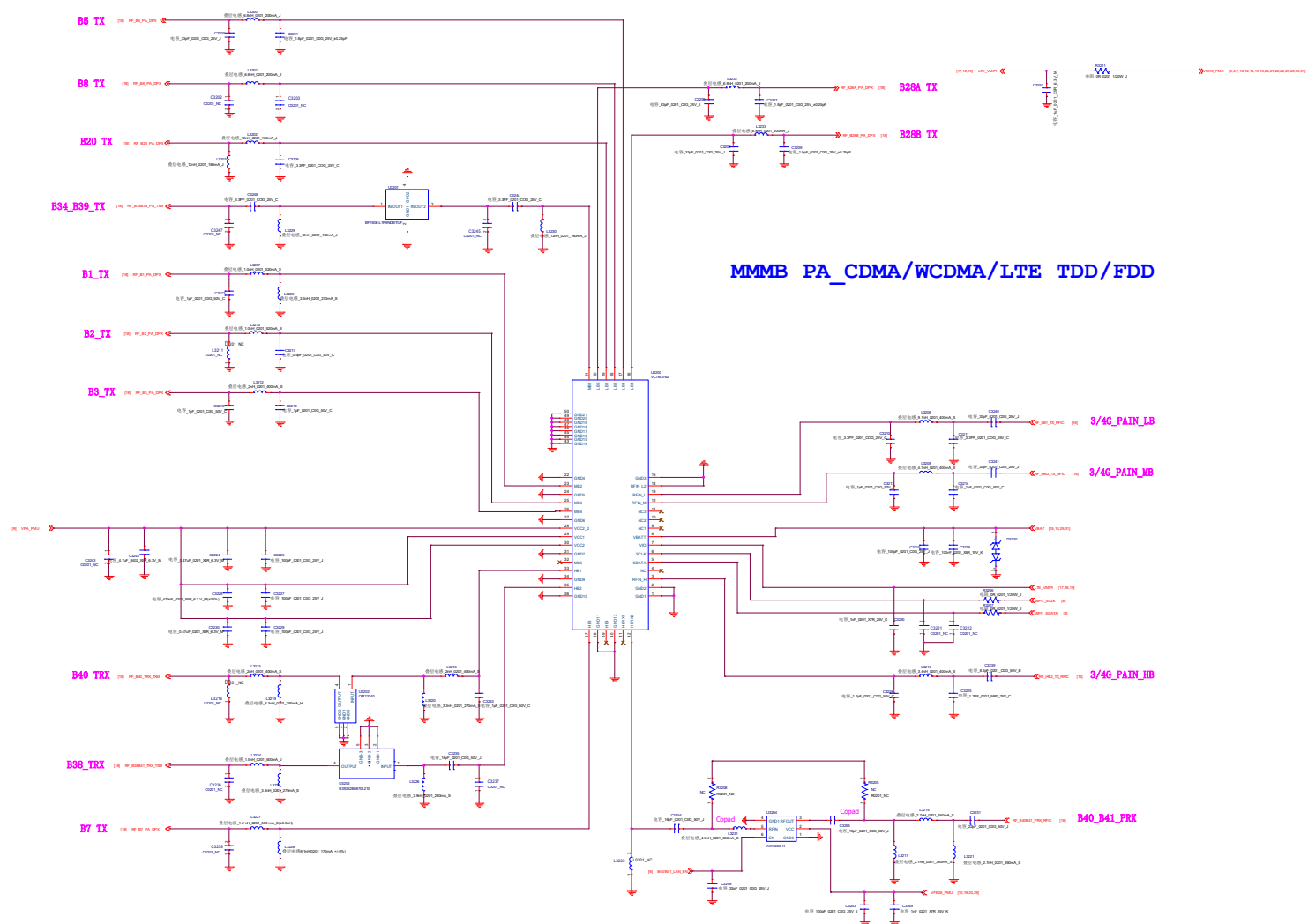
Power For MT6177



GND

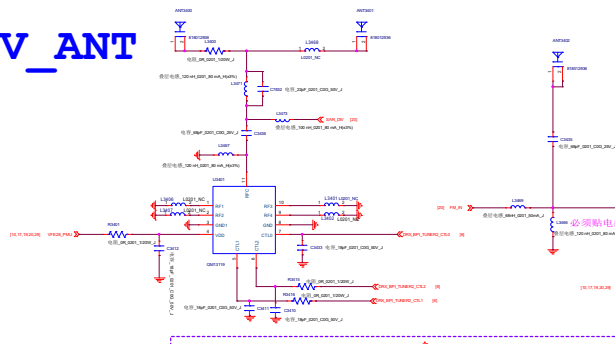


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Title 31_RF_MT6177_Pin_Out			
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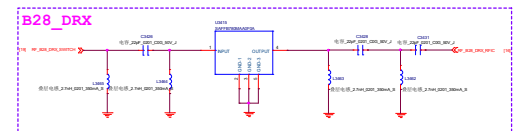
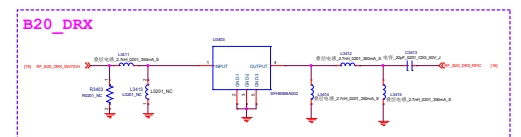
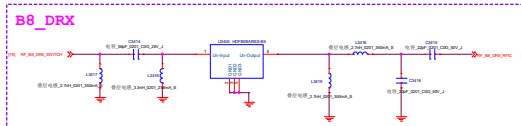
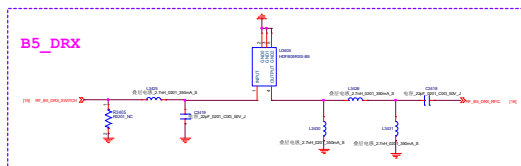


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Title 32_RF_MT6177_RF_TX		
Size E	Project F9_MB_V1	Rev v1
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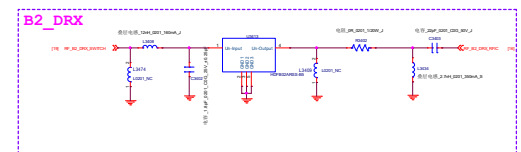
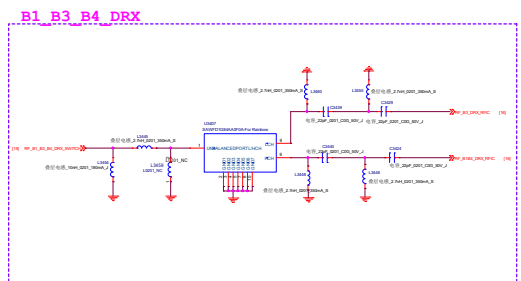
DIV_ANT



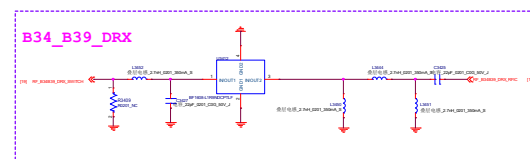
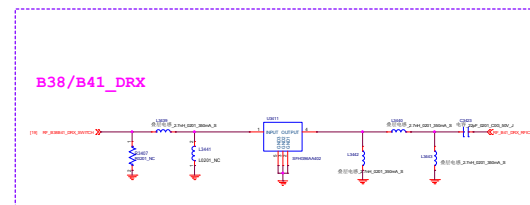
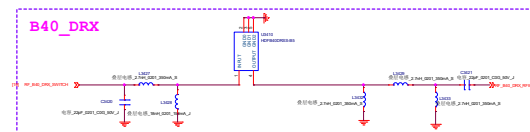
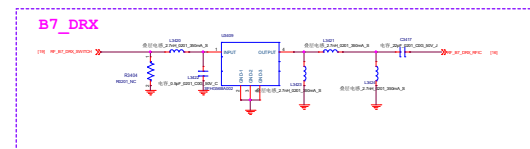
DIV_LB



DIV_MB



DIV_HB



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Title 34_RF_MT6177_RF_DRX

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同轴座

空口检测

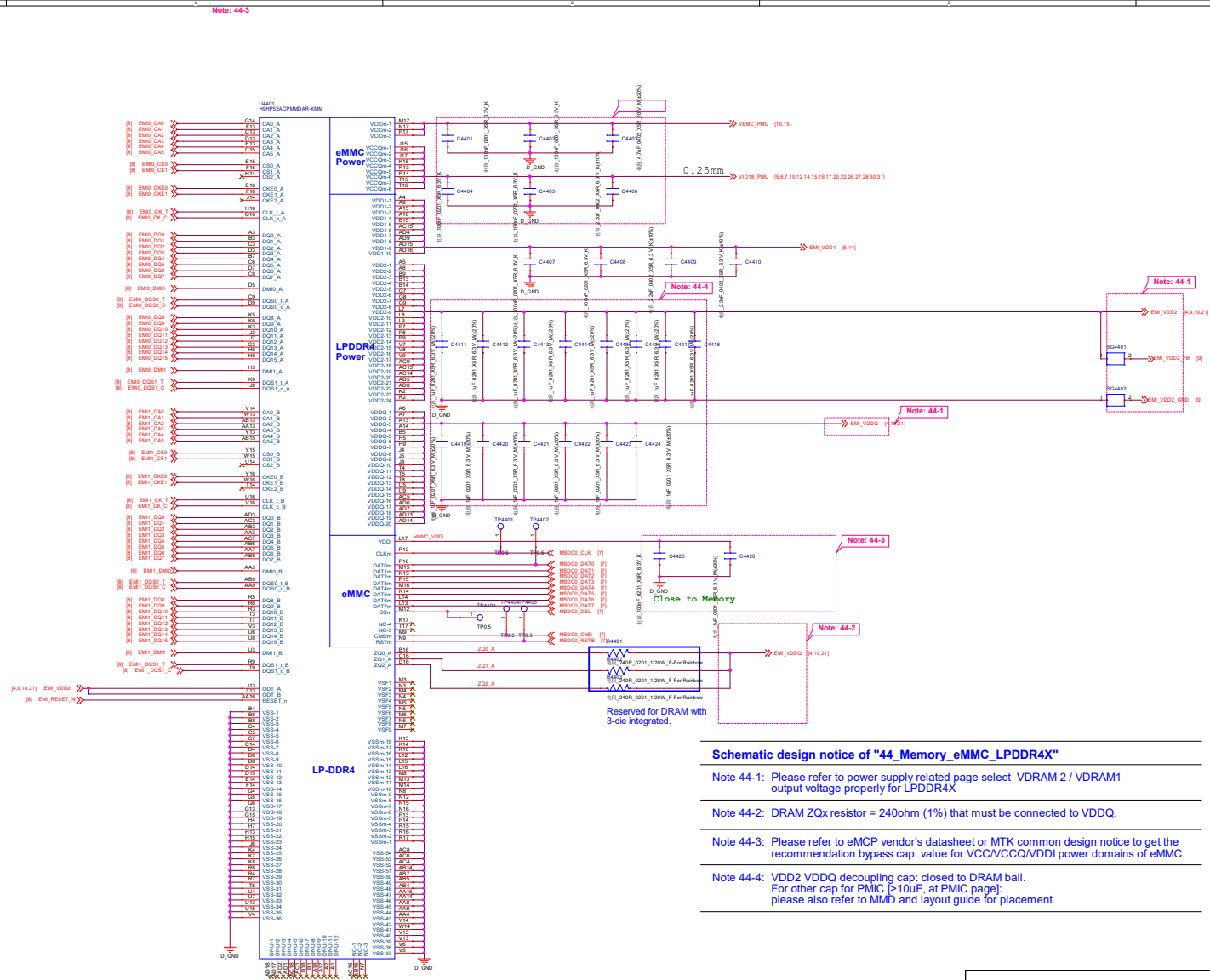
TAS

MAIN DPDT SWITCH

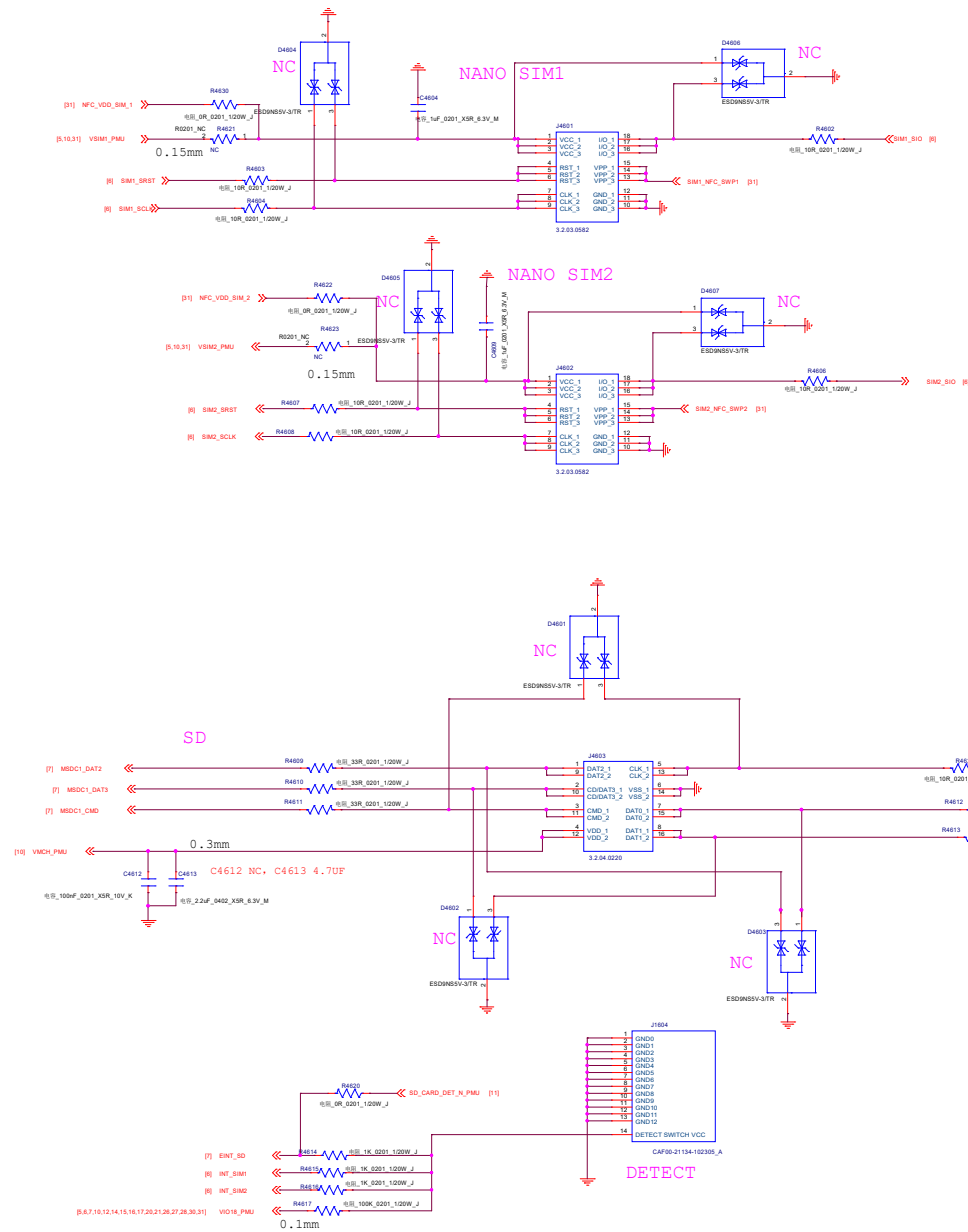
SAR Sensor

FM

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Note 46-1:

Schematic design notice of "46_MEMORY_SD Card" page.

For better ESD performance, please select suitable components for system protection.

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Title 01_Block_Diagram

Size D

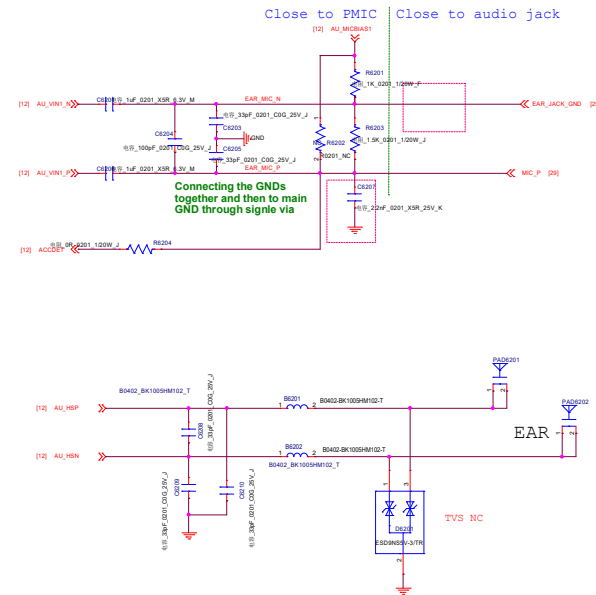
Project J15_MB_V1

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Earphone Microphone



Schematic design notice of "62_PERI_AUDIO_IO" page.

Note 62-1: Part # of BEAD6202, BEAD6203, BEAD6204 and BEAD6205 needs changed to "BLM18BD102SN1" for high THD performance (~90dB) but this BOM change will results in FM RSSI 10dB degraded .

Note 62-2: Reserved Cap for CS/RS test, please double check multi-key function when used

Note 62-3:

Earphone Jack	Earphone Jack
0 ohm	100nF

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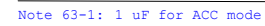
Title 01_Block_Diagram

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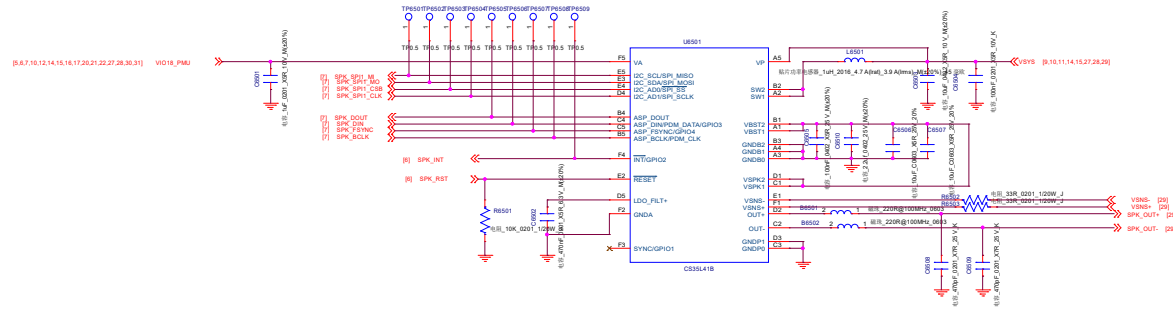
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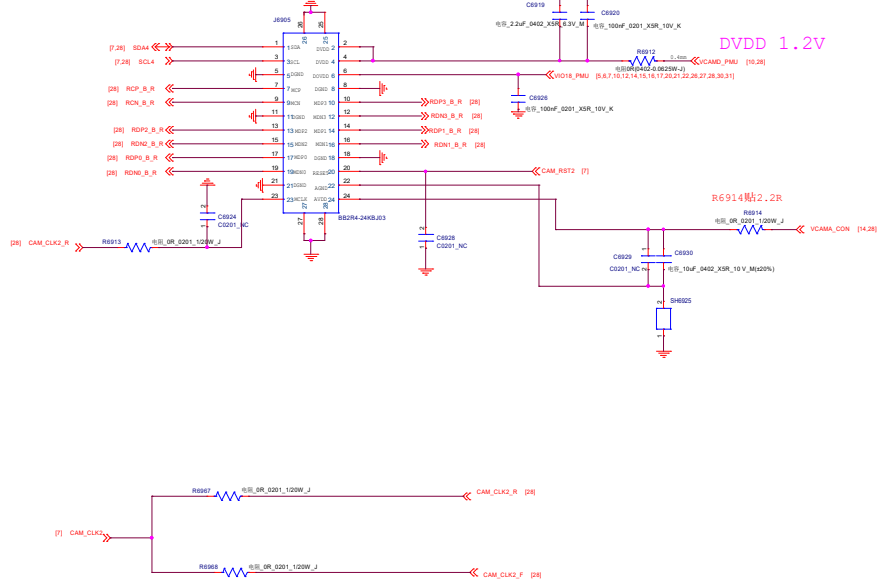
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
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Speaker Amplifier

MT6660 Speaker AMP I2C Address: 0x34 (Write:0x68, Read:0x69) when AD81 = GND; AD82 = GND.

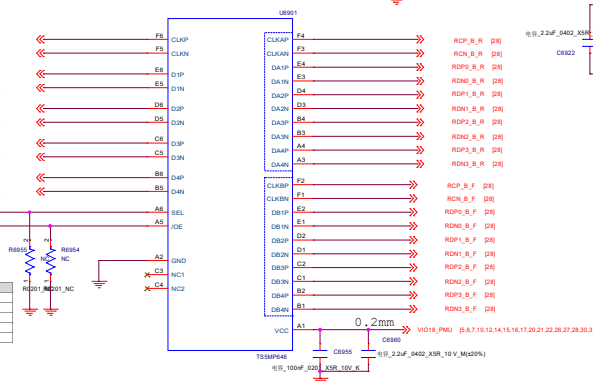


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[illegible]

INPUTS			OUTPUTS	
CLK_REG1	CLK_REG2	MCLK_IN	CLK_OUT1	CLK_OUT2
L	L	X	L	L
L	H	CLK	L	CLK
H	L	CLK	CLK	L
H	H	CLK	CLK	CLK

OE	BEL	Function
H	X	IO pins High-impedance
L	L	$CLK(PIN) = CLK(A/PIN)$ $DIO(PIN) = DIO(A/PIN)$
L	H	$CLK(PIN) = CLK(B/PIN)$ $DIO(PIN) = DIO(B/PIN)$

[illegible]

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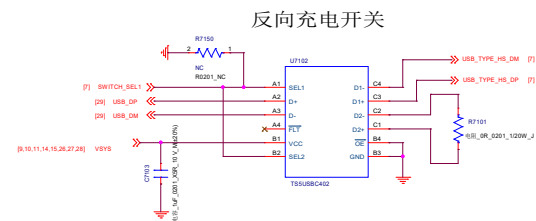
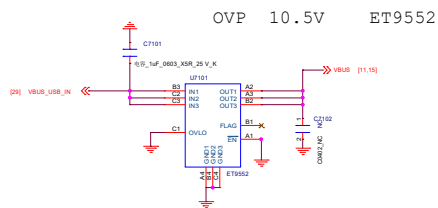
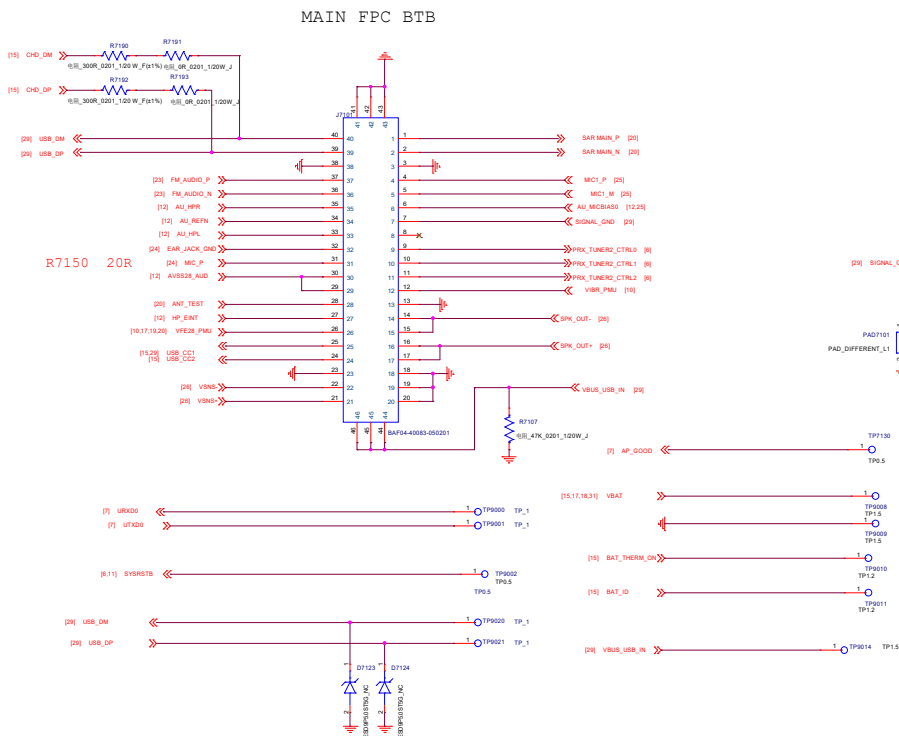
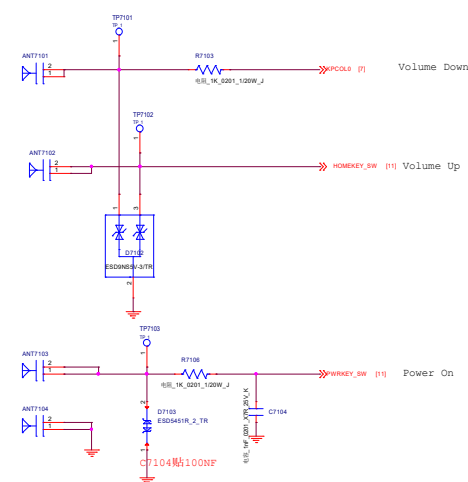


Table 1. Function Table

OE	SEL1	SEL2	D- Connection	D+ Connection
H	X	X	High-Z	High-Z
L	L	L	D- to D1-	D+ to D1+
L	L	H	D- to D1-	D+ to D2+
L	H	L	D- to D2-	D+ to D1+
L	H	H	D- to D2-	D+ to D2+



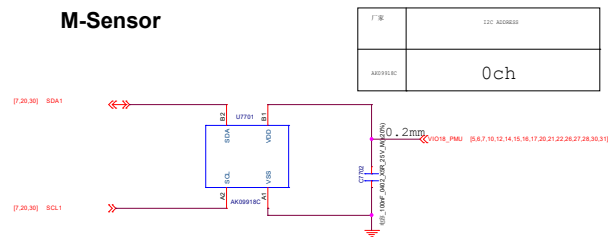
KEY BTB



USB

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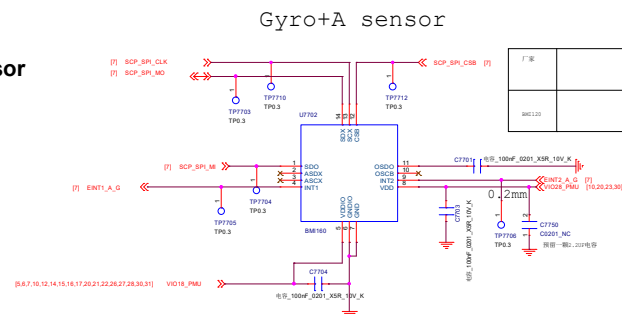
M-Sensor



厂家	12C: AK8963
地址/ID	0ch

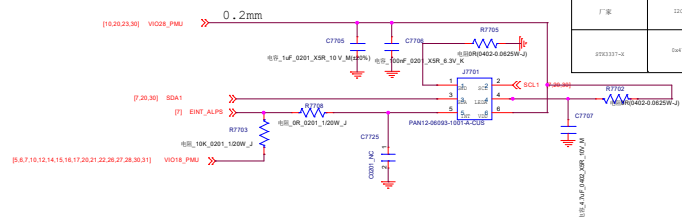
Accerometer + Gyro Sensor

default SPI (if sensor hub support)



厂家	12C: ADXRS450
地址/ID	0x68

ALS PS STK3337-X



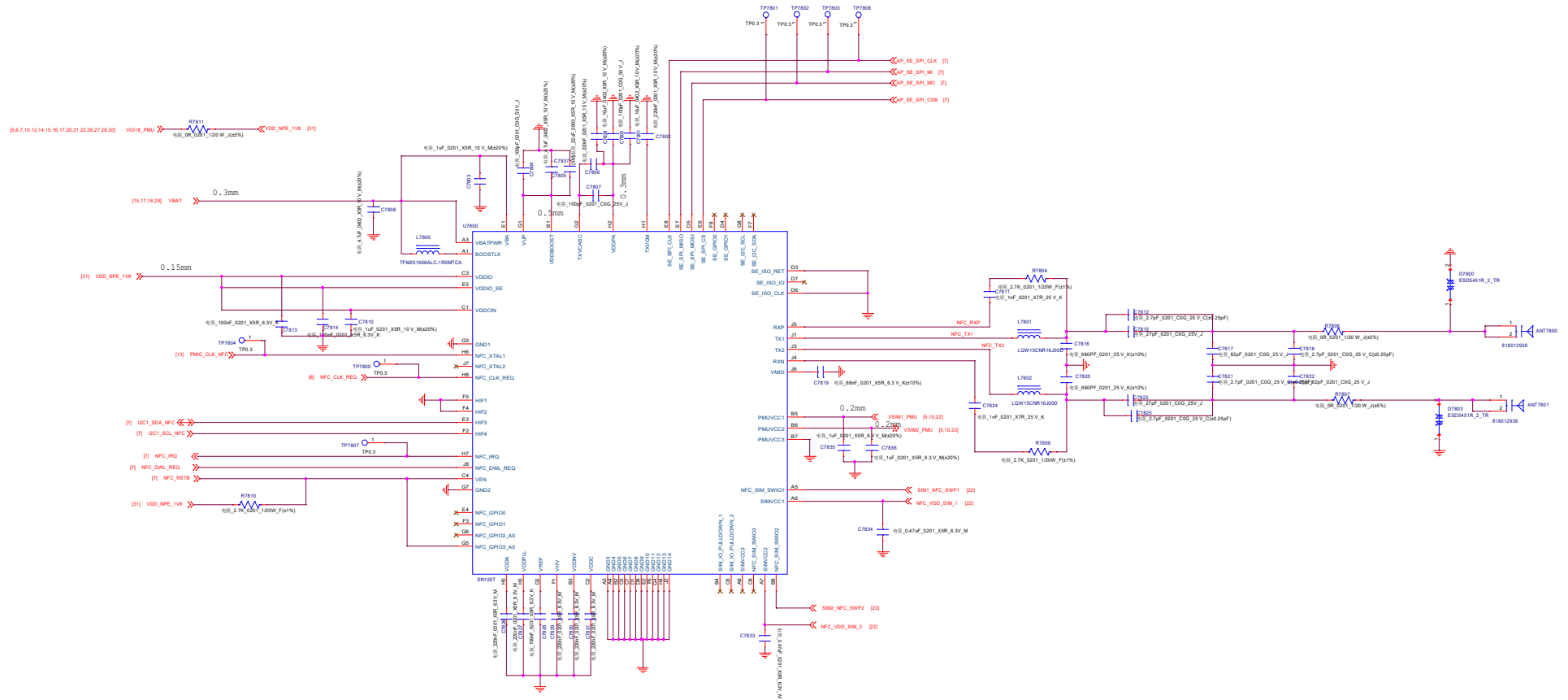
厂家	12C: ADXRS450
地址/ID	0x68

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